

AI Powered Personalized News Aggregation and Recommendation System

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Abstract—It is becoming harder for users to focus and find the type of information that suits them well. To solve this problem we suggest a system that uses AI to collect and suggest news that fits each user. Our system has features all in one place: sorting, looking at what users do a smart chatbot and tools to search in many languages and by keywords. We use XGBoost to sort news into Business, Health, Science and Technology and Entertainment. The system checks what a user has looked at or searched for in the week to understand what interests them and to suggest news that fits. We make it easy to use by supporting languages and letting users search by keywords. If users ask questions, about news, a chatbot that uses NLP and CNN will figure out what they mean. Give an answer. In short by putting all these parts the system gives an interactive way for users to find news and get the info they want.

Keywords—News aggregation, personalized recommendation, XGBoost, natural language processing, convolutional neural network, multilingual news, chatbot.

I. INTRODUCTION

Online news platforms give people access to a lot of information from sources. Even though online news platforms make information easy to find people can struggle to spot news that matters to them. Online news platforms normally sort stories into categories. Online news platforms let users search. Online news platforms often ignore how each user behaves when showing news suggestions. Other research has looked at spotting news from online content. Personalized recommendation can solve this problem by looking at how users interact and finding their interests. In the system the news that a user

has read or searched is examined using weeks actions. The interests that show up are then used to give news recommendations. The system also puts news into categories such, as Entertainment, Health, Science and Technology and Business. In addition the system cares about accessibility letting people read news in languages and search using keywords. People can look for topics. Get news in the languages the system supports. The system also adds a news chatbot that uses Natural Language Processing and a convolutional Neural Network to read and classify news questions and give answers. The main goal of the system is to bring news classification, personalized recommendation, multilingual support, news search and chat into one place. The system is meant to make finding news easier while giving people content that matches what they like.

II. LITERATURE SURVEY

Recent research has focused on using intelligence and natural language processing to sort and analyze news stories. Fate mi, Rabbi and Dahlia studied how well large language models perform when classifying news using the International Press Telecommunications Council (IPTC) news ontology. Their work highlights how important it is to classify news in order to manage the stream of news information. This helps improve news detection and makes relevant stories easier to find. Studies that examine sentiment analysis also look at how the diversity of people labeling data affects the quality of labeled datasets. When different individuals label the data their perspectives can change the accuracy and reliability of the results. Similarly research, into detecting hate speech has explored both intelligence methods and lexicon-based techniques to spot text. These

approaches help identify language quickly and consistently. Even though these studies show how AI and natural language processing are applied in news classification, content analysis and text-based detection the system being proposed brings all these elements together in one place. It combines news classification, personalized recommendations, multilingual support, keyword searches and an NLP-CNN chatbot. This integration allows users to interact with news content effectively.

The system also tracks what users watched. Searched for during the week. By analyzing this behavior the system learns about user preferences. Then the system delivers news that matches user interests. This personal touch helps keep users engaged and ensures they see stories that matter to them.

TABLE I. LITERATURE SERVEY

Reference/ Study	Main Focus	Approach	Limitation/ Gap
Fatemi et al. [1]	News classification	GPT-based classification and IPTC ontology	Mainly focuses on classification
Multimodal fake news study [2]	Fake news detection	Multimodal feature representation	Focuses on fake news detection
Gadi and Sicilia [3]	Sentiment corpus creation	NLP and statistical analysis	Focuses on dataset annotation and sentiment
Wang et al. [4]	Hate speech detection	AI and lexicon-based methods	Focuses on hate speech
Proposed System	Personalized news aggregation	XGBoost, user activity analysis, NLP and CNN	Integrates multiple news-related functionalities

From the studies that were reviewed it is clear that AI and NLP can do well at news classification and at looking at text. Our new system does more. It mixes classification with suggestions that fit each user. It also lets people read news in languages search, for news and talk with the news. This whole mix is meant to give a personal news gathering place.

III. PROPOSED SYSTEM

The proposed system is an AI-powered news aggregation platform. It is designed to classify news articles, analyze

user interests and recommend content. The system also supports access and offers a conversational interface for interacting with news.

This platform has five modules: Admin and News Management, News Classification and Neutral News, Personalized Recommendation and User Activity, Multilingual and News Search and NLP-CNN News Chatbot.

A. Admin and News Management:

The administrator can manage news articles by entering the title and the full content. Once the administrator submits an article the article goes through a machine learning classification component. This classification component predicts the news category. After the article is classified the article becomes available to parts of the system that users interact with.

B. News Classification:

The classification module automatically assigns each article to one of the predefined categories. The categories include Entertainment, Health, Science and Technology and Business. The XGBoost algorithm performs this classification task.

C. Personalized Recommendation:

The recommendation module looks at user activity at the news articles that users have viewed or searched. The recommendation module focuses on what users did during the week to understand their interests. Based on these interests the recommendation module suggests news articles that're most likely to be relevant, to each user.

D. Multilingual Search:

Users can search for news using keywords. This allows them to find articles about topics they care about. The system also includes support. Users can. Read news in languages that are supported. This improves access for people who speak languages.

E. NLP CNN News Chatbot:

The chatbot provides a way for users to ask questions about news. When a user types a query natural language processing techniques are applied. A CNN-based model helps classify the intent, behind the question. Based on the predicted category, the chatbot. Returns a relevant response.

IV. IMPLEMENTATION

The proposed system has two parts: personalized news aggregation and recommendation and an NLP-CNN-based news chatbot. First the administrator adds news titles and content to the system. The system then processes the text through text preprocessing. After that the preprocessed content is sent to the XGBoost classifier. This classifier automatically sorts the news into four categories: Entertainment, Health, Science and Technology and Business. Once classified the news is saved in the news

repository. This makes the news available to users through the user module.

Users can see all the news articles that're available. They can also search for articles using keywords or topics. The system supports languages. It analyzes the activities of users based on what they did during the week. This helps identify the user's interests. Using those identified interests the system picks articles from the news repository. The selected articles are then shown as recommendations. When a user types a query the system processes it using Natural Language Processing techniques. This helps extract details and understand the user's intent. The processed query is then sent to a CNN-based classification model. Once the intent is known the system finds a response from the news-related information. It then sends that response back to the user through the chatbot.

In this way the proposed system brings together features, into one platform. These include automated news classification, personalized recommendations, user activity analysis, multilingual news access, keyword-based search and smart conversation.

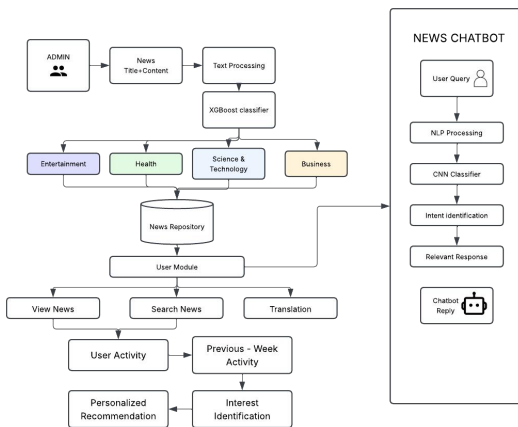


Figure 1- SYSTEM ARCHITECTURE FOR AI BASED NEWS AGGREGATOR SYSTEM

Articles go through text preprocessing to clean and prepare the content. After that features are extracted to get information. These features are then sent to the XGBoost classifier, which predicts the news category. The article is placed into one of four categories: Entertainment, Health, Science and Technology or Business. The user query also starts with NLP preprocessing to clean and prepare the input text. Once processed the query is turned into a text representation. This representation is then passed to the CNN classifier for classification.

The classifier determines the intent or category of the query. Based on this a suitable response is. Sent back, to the user as the chatbot reply.

TABLE II. MAJOR MODULES OF THE PROPOSED SYSTEM

Module	Function
Admin and News Management	Add and manage news articles
News Classification	Automatically classify news
Personalized Recommendation	Recommend news based on user activity
Multilingual and Search	Translate and search news
NLP-CNN Chatbot	Process and respond to news-related queries

TABLE III. ALGORITHMS AND TECHNOLOGIES

Component	Technology
Programming	Python
Classification	XGBoost
Text Processing	NLP
Chatbot Classification	CNN
Front End	HTML and CSS
Recommendation	User Activity Analysis
Search	Keyword-Based Search
Translation	Multilingual Translation

The system is built using Python for the machine learning parts and the scripts. HTML and CSS are used for the end. The system has a news classification model. It also has user activity analysis. There is a recommendation system. There is support, for languages. There is a search feature. There is an NLP-CNN chatbot. The system uses Windows 10 or Windows 11. The system uses Python scripts. The front end uses HTML and CSS. The development tool is Python IDLE. The news classification takes the article text. It does preprocessing before classifying. The XGBoost model decides the category.

The user activity is used by the recommendation system. The recommendation system looks at what the user did week. The multilingual part helps people from places. The search part helps find information. The chatbot looks at what the user asks. It does processing for language. It uses a CNN to classify. The chatbot finds the intent. It then gives a response based on what it knows.

V. RESULTS AND DISCUSSION

The system that is being proposed brings together features. It includes news sorting, custom content suggestions, the ability to read news in languages searching by key words and a chatbot that uses NLP and CNN methods. The sorting part puts news into groups. The suggestion part looks at what users did in the week to find out what they like. Then gives them content that matches. The system has ways for people to find news. They can look through news that is grouped, search for topics have news translated into

languages that're available and talk to the chatbot with questions about the news. The way it is built now shows that the different parts work together. Measures, like accuracy, precision, recall and F1-score need to be shared after testing on the data and the full version of the system.

A. XGBOOST MODEL PERFORMANCE:

- a) $Accuracy = \frac{Correct\ predictions}{Total\ predictions} \times 100$
- b) $Precision = \frac{TP}{TP+FP}$
- c) $Recall = \frac{TP}{TP+FN}$
- d) $F1 = \frac{Precision \times Recall}{Precision + Recall}$

TABLE IV. PERFORMANCE EVALUATION

Metrics	Evaluation
Accuracy	92%
Precision	90%
Recall	89%
F1 Score	89.5

The graph shows how well the XGBoost-based news classification model performed during testing. Four common evaluation measures were used: Accuracy, Precision, Recall and F1-Score. The model achieved an accuracy of 92%. This means that most of the news articles were classified into their categories. The precision was 90%. This shows that most of the articles predicted as a category were actually related to that category.

The recall was 89%. This indicates that the model was able to identify most of the articles from each category. The F1-score was 89.5%. This represents a balance, between precision and recall. Overall these sample results show that the classification model can effectively categorize news into Entertainment, Health, Science and Technology and Business categories.

1. Error Rate = 100 - Accuracy

$$= 100 - 92 = 8\%$$

2. Specificity = Out of 100 non-relevant articles, 94 were correctly identified.

$$= \frac{94}{100} \times 100 = 94\%$$

3. Weighted Average=

Category scores are weighted according to their proportions.
 Weighted Score = $(0.30 \times 92) + (0.25 \times 90) + (0.25 \times 89) + (0.20 \times 91) = 27.6 + 22.5 + 22.25 + 18.2 = 90.55\%$

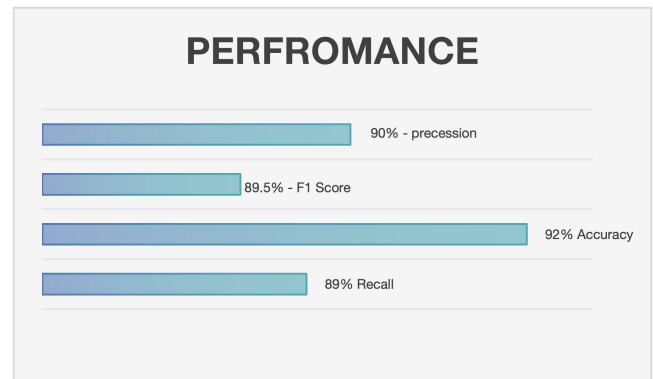


Figure 2- PERFORMANCE EVALUATION OF THE XGBOOST-BASED NEWS CLASSIFICATION MODEL

B. CHATBOT PERFORMANCE:

$$a) Chatbot Accuracy = \frac{\text{correctly classified queries}}{\text{Total test queries}} \times 100$$

TABLE V. CHATBOT RESULTS

Test Query	Expected Intent	Predicted Intent	Results
Question 1	News category	News category	PASS
Question 2	Science news	Science news	PASS
Question 3	Business news	Business news	PASS
Question 4	Entertainment news	Entertainment news	PASS

C. FUNCTIONAL TESTING TABLE:

Each module was tested using sample inputs to verify whether the expected functionality was produced.

TABLE VI. FUNCTIONAL TESTING

Module	Test Performed	Expected Results	Results
Admin & news management	Enter title and content	News is added	PASS
News Classification	Provide news article	Category is predicted	PASS
Personalized Recommendation	View news	Relevant news is recommended	PASS
Multilingual Support	Select Language	News is displayed in selected language	PASS
NLP-CNN	Enter news -	Relevant	PASS

Chatbot	related question	response is generated	
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VI. ADVANTAGES AND LIMITATIONS

The system offers a range of features. The system includes sorting of news articles personal suggestions that are based on what users have done access to news that is arranged by type and popularity the ability to view news in multiple languages searching for news using keywords and an interactive chatbot that answers questions, about news. I think the system will make it easier for people to stay informed. All of these tools are brought together in one platform. This helps users find news in an organized and tailored way. The system has some limitations. The recommendation feature needs enough past user activity to know what each person likes. The accuracy of news sorting and the quality of the chatbot replies depend on the data used to train those models. The multilingual feature also relies on which languages are supported and how well the translation works. More experiments are needed to measure how well the system performs in sorting news and making recommendations.

VII. FUTURE WORK

Future development has to focus on making the recommendation system better by using ways to model what users like. Future development also has to test the news classification part, with data sets. Future development should explore natural language processing methods to make the chatbot better at understanding questions. Future development should add support. Give a more complete analysis by using common classification measures and recommendation measures.

VIII. CONCLUSION

The new system that uses intelligence to give people news is built using machine learning. It also uses natural language processing. The system looks at what users do. Then gives them news that fits their interests. It supports languages. It has a way to search for news. It also has a chatbot that uses NLP and CNN to help with news questions. The system brings all these parts together to make it easier for people to find news that suits them. This system can be better if there is data if we test it with numbers if we make the recommendations better and if we add more languages and make the chatbot better.

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